measurement on the right-moving particle, so, in short, the left-moving particle couldn't care less what you do. Thus, any feature that you actually learn or could in principle learn about the left-moving particle from studying its right-moving counterpart must be a *definite*, *existing* feature of the left-moving particle, totally independent of your measurement. And since if you had measured the position of the right particle you would have learned the position of the left particle, and if you had measured the velocity of the right particle you would have learned the velocity of the left particle, it must be that the left-moving particle actually has both a definite position and velocity. Of course, this whole discussion could be carried out interchanging the roles of left-moving and right-moving particles (and, in fact, before doing any measurement we can't even say which particle is moving left and which is moving right); this leads to the conclusion that both particles have definite positions and speeds.

Thus, EPR concluded that quantum mechanics is an incomplete description of reality. Particles have definite positions and speeds, but the quantum mechanical uncertainty principle shows that these features of reality are beyond the bounds of what the theory can handle. If, in agreement with these and most other physicists, you believe that a full theory of nature should describe every attribute of reality, the failure of quantum mechanics to describe both the positions and the velocities of particles means that it misses some attributes and is therefore not a complete theory; it is not the final word. That is what Einstein, Podolsky, and Rosen vigorously argued.

The Quantum Response

While EPR concluded that each particle has a definite position and velocity at any given moment, notice that if you follow their procedure you will fall short of actually determining these attributes. I said, above, that you could have chosen to measure the right-moving particle's velocity. Had you done so, you would have disturbed its position; on the other hand, had you chosen to measure its position you would have disturbed its velocity. If you don't have both of these attributes of the right-moving particle in hand, you don't have them for the left-moving particle either. Thus, *there is no conflict with the uncertainty principle:*